Unit 4: Autonomous Equations, Equilibrium Solutions, and Stability

Prepared by Staples Education

Practice Set: Problems and Complete Solutions

Part II — Review Guide: Practice Problems

Work the following ten problems in order. They climb from locating equilibria, through stability tests and phase lines, into the logistic model, and finish with the Bernoulli substitution $v = y^{1-n}$ and a full solve. Solutions follow in Part III.

1. Find all equilibrium solutions of $\dfrac{dy}{dt} = y(y-3)$.
2. Classify the equilibrium $y = 2$ of $\dfrac{dy}{dt} = y - 2$ using the sign of f'.
3. For $\dfrac{dy}{dt} = y - y^3$, find every equilibrium and classify each with the linearized test.
4. Classify the equilibrium $y = 1$ of $\dfrac{dy}{dt} = (y-1)^2$. Why does the f' test alone not settle it?
5. Build the phase line for $\dfrac{dy}{dt} = y^2 - 4$: mark the equilibria, set the arrows, and classify each.
6. For the logistic equation $\dfrac{dP}{dt} = rP\left(1 - \dfrac{P}{K}\right)$ with $r > 0$, find both equilibria, classify them, and give the population of fastest growth.
7. With $r = 0.5$ and $K = 100$, compute $\dfrac{dP}{dt}$ at $P = 25$ and state the population at which the growth rate is greatest.
8. For $y' + y = xy^3$, identify n, write the substitution v, and compute $\dfrac{dv}{dx}$ in terms of y'.
9. Reduce $y' + \dfrac{1}{x}\, y = x\, y^2$ to a first-order *linear* equation in $v = y^{1-n}$ (do not solve it yet).
10. **(Capstone)** Solve the Bernoulli equation $y' + y = y^2$ completely: reduce it with $v = y^{-1}$, solve the linear equation, and back-substitute for y.

Part III — Complete Solutions

Solution 1. Equilibria are the roots of $f(y) = 0$:

$$y(y-3) = 0 \implies y = 0 \text{ and } y = 3.$$

Solution 2. Here $f(y) = y - 2$, so $f'(y) = 1$:

$$f'(2) = 1 > 0 \implies \text{unstable (a source).}$$

A positive slope of f at the equilibrium repels nearby solutions.

Solution 3. Factor for the equilibria, then apply $f'(y) = 1 - 3y^2$:

$$y - y^3 = y(1-y)(1+y) = 0 \implies y = -1,\ 0,\ 1.$$

$$f'(0) = 1 > 0 \text{ (unstable)}, \qquad f'(\pm 1) = 1 - 3 = -2 < 0 \text{ (stable)}.$$

So $y = \pm 1$ are sinks and $y = 0$ is a source.

Solution 4. The only equilibrium is $y = 1$. The linearized test is inconclusive because

$$f'(y) = 2(y-1) \implies f'(1) = 0.$$

Fall back on the sign of $f(y) = (y-1)^2$, which is ≥ 0 everywhere and *never* changes sign across $y = 1$:

$$y < 1:\ f > 0\ (\uparrow), \qquad y > 1:\ f > 0\ (\uparrow).$$

Solutions rise toward 1 from below but away from 1 above, so $y = 1$ is **semi-stable**.

Solution 5. Equilibria where the rate vanishes:

$$y^2 - 4 = 0 \implies y = -2,\ 2.$$

Test the rate sign on each interval:

$$y < -2:\ (+)\uparrow, \qquad -2 < y < 2:\ (-)\downarrow, \qquad y > 2:\ (+)\uparrow.$$

Arrows converge on -2 (**stable**) and diverge from 2 (**unstable**).

Solution 6. Set the product to zero:

$$rP\left(1 - \frac{P}{K}\right) = 0 \implies P = 0 \text{ and } P = K.$$

Classify with $f'(P) = r\left(1 - \dfrac{2P}{K}\right)$:

$$f'(0) = r > 0 \text{ (unstable)}, \qquad f'(K) = r(1-2) = -r < 0 \text{ (stable)}.$$

The growth rate, a downward parabola in P with roots 0 and K, peaks at the midpoint:

$$P = \frac{K}{2}.$$

Solution 7. Substitute into $rP\left(1-\frac{P}{K}\right)$:

$$\frac{dP}{dt} = 0.5 \cdot 25\left(1 - \frac{25}{100}\right) = 12.5 \cdot 0.75 = 9.375.$$

The greatest growth occurs at $P = \frac{K}{2} = 50$.

Solution 8. The right side is y^3, so $n = 3$ and the linearizing exponent is $1 - n = -2$:

$$v = y^{1-n} = y^{-2}, \qquad \frac{dv}{dx} = (1-n)y^{-n}\frac{dy}{dx} = -2\,y^{-3}\,y'.$$

Solution 9. Here $n = 2$, so $v = y^{-1}$ and $v' = -y^{-2}y'$. Divide every term by y^2:

$$y^{-2}y' + \frac{1}{x}\,y^{-1} = x.$$

Replace $y^{-1} = v$ and $y^{-2}y' = -v'$:

$$-v' + \frac{1}{x}\,v = x \implies v' - \frac{1}{x}\,v = -x,$$

a first-order linear equation in v.

Solution 10. With $n = 2$, set $v = y^{-1}$, $v' = -y^{-2}y'$. Divide $y' + y = y^2$ by y^2:

$$y^{-2}y' + y^{-1} = 1.$$

Substitute $y^{-1} = v$, $y^{-2}y' = -v'$:

$$-v' + v = 1 \implies v' - v = -1.$$

Integrating factor $\mu = e^{-x}$:

$$\left(e^{-x}v\right)' = -e^{-x} \implies e^{-x}v = e^{-x} + C.$$

So $v = 1 + Ce^{x}$, and back-substituting $v = y^{-1}$:

$$y = \frac{1}{v} = \frac{1}{1 + Ce^{x}}.$$

Unit 4 Key Takeaway

Stability is settled by two signs: $f(c) = 0$ locates an equilibrium and the sign of $f'(c)$ (or, when that is zero, the sign of f on each side) classifies it. The logistic equation packages this with a stable carrying capacity K and fastest growth at $\frac{K}{2}$. Every Bernoulli equation yields to $v = y^{1-n}$, which trades the lone power of y for a clean linear solve.